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CORRECTED MAGNETIC PROXIMITY FUZE SIMULATION

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Starting corrected simulation...
Torpedo approaching target...

Time: 1.4 s, Distance: 572.5 m, Ship B: 0.00e+00 T, Det: 2.25e-04, Der: 0.00e+00
Time: 2.8 s, Distance: 544.6 m, Ship B: 0.00e+00 T, Det: 1.10e-04, Der: 0.00e+00
Time: 4.2 s, Distance: 516.6 m, Ship B: 0.00e+00 T, Det: 1.85e-04, Der: 0.00e+00
Time: 5.6 s, Distance: 488.6 m, Ship B: 0.00e+00 T, Det: 1.07e-04, Der: 0.00e+00
Time: 7.0 s, Distance: 460.6 m, Ship B: 0.00e+00 T, Det: 2.18e-04, Der: 0.00e+00
Time: 8.4 s, Distance: 432.7 m, Ship B: 0.00e+00 T, Det: 2.91e-04, Der: 0.00e+00
Time: 9.8 s, Distance: 404.7 m, Ship B: 0.00e+00 T, Det: 1.17e-04, Der: 0.00e+00
Time: 11.2 s, Distance: 376.7 m, Ship B: 0.00e+00 T, Det: 2.26e-04, Der: 0.00e+00
Time: 12.6 s, Distance: 348.8 m, Ship B: 0.00e+00 T, Det: 8.95e-05, Der: 0.00e+00
Time: 14.0 s, Distance: 320.8 m, Ship B: 0.00e+00 T, Det: 1.91e-04, Der: 0.00e+00
Time: 15.4 s, Distance: 292.9 m, Ship B: 0.00e+00 T, Det: 5.58e-05, Der: 0.00e+00
Time: 16.8 s, Distance: 265.0 m, Ship B: 0.00e+00 T, Det: 8.98e-05, Der: 0.00e+00
Time: 18.2 s, Distance: 237.0 m, Ship B: 4.58e-08 T, Det: 4.65e-03, Der: 5.72e-04
Signal efficiency: 101.5% (Expected: 0.005, Actual: 0.004653)
Time: 19.6 s, Distance: 209.2 m, Ship B: 6.67e-08 T, Det: 6.57e-03, Der: 9.41e-04
Signal efficiency: 98.5% (Expected: 0.007, Actual: 0.006571)
Time: 21.0 s, Distance: 181.3 m, Ship B: 1.05e-07 T, Det: 1.04e-02, Der: 1.79e-03
Signal efficiency: 98.9% (Expected: 0.010, Actual: 0.010384)
Time: 22.4 s, Distance: 153.5 m, Ship B: 1.79e-07 T, Det: 1.79e-02, Der: 3.77e-03

Signal efficiency: 99.6% (Expected: 0.018, Actual: 0.017864)
Time: 23.8 s, Distance: 125.8 m, Ship B: 3.56e-07 T, Det: 3.50e-02, Der: 9.81e-03
Signal efficiency: 98.3% (Expected: 0.036, Actual: 0.034953)

*** DETONATION TRIGGERED ***

Time: 24.09 seconds
Distance to target: 119.9 meters
Total magnetic field: 5.04e-05 Tesla
Ship magnetic contribution: 4.23e-07 Tesla
Detection signal: 4.19e-02 (threshold: 1.00e-04)
Derivative signal: 1.25e-02 (threshold: 2.50e-04)
Magnitude detection: 1, Derivative detection: 1

=== SIGNAL PROCESSING DEBUG ===

Raw sensor: 5.04e-05 T
After DC removal: 4.27e-07 T
Amplified signal: 0.042656
Filtered signal: 0.041944
Expected vs Actual ratio: 99.1%

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CORRECTED SIMULATION RESULTS

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Minimum approach distance: 119.9 meters at 24.1 seconds

*** DETONATION SUCCESSFUL ***

Detonation occurred at: 24.1 seconds

Detonation distance: 119.9 meters

Fuze effectiveness: SUCCESSFUL

=== PERFORMANCE ANALYSIS ===

Maximum detection signal: 4.19e-02

Maximum ship magnetic field: 4.23e-07 Tesla

Detection method: MAGNITUDE (Primary)

Maximum magnetic field deviation: 5.00e-05 Tesla

Maximum ship magnetic field: 4.23e-07 Tesla

Corrected simulation completed successfully!

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Corrected Magnetic Proximity Fuze Simulation Results

